

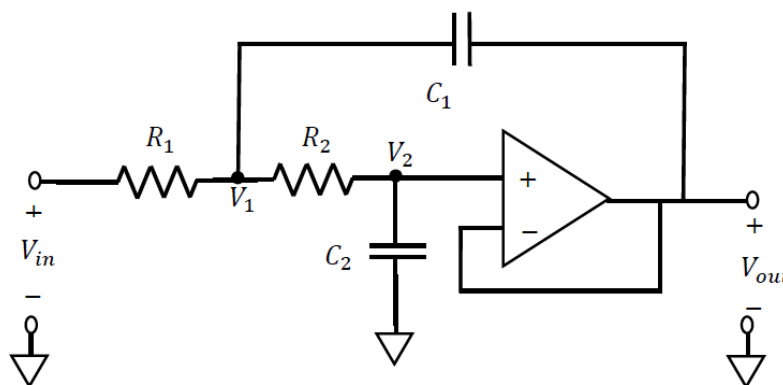
Lab 8 - Transient Response II

Objective

The goal of this lab is to explore the transient behavior of a 2nd order circuit. To do so, you will build an op-amp circuit and measure the transient (step) response of the circuit and compare with theory and simulation results.

Introduction

Textbooks often discuss using circuits with resistors, capacitors and inductors to produce 2nd order responses. In practice, we tend to avoid using circuits with inductors as they are not easily integrated onto a chip and therefore result in circuits that cannot easily be miniaturized. In this lab we will use an op-amp circuit with resistors and capacitors that can synthesize any desired 2nd order response. The circuit shown below is known as a Sallen-Key circuit.



First, do a KCL analysis at node V_1 , namely:

$$\frac{V_{in} - V_1}{R_1} + \frac{V_{out} - V_1}{R_2} + C_1 \frac{d}{dt}(V_{out} - V_1) = 0 \Rightarrow$$

$$V_{in} = V_{out} + \left(1 + \frac{R_1}{R_2}\right)(V_1 - V_{out}) + R_1 C_1 \frac{d}{dt}(V_1 - V_{out})$$

Next, do a KCL analysis at node V_2 , namely:

$$\frac{V_1 - V_{out}}{R_2} = C_2 \frac{dV_{out}}{dt} \Rightarrow$$

$$V_1 - V_{out} = R_2 C_2 \frac{dV_{out}}{dt} \Rightarrow$$

$$\frac{d}{dt}(V_1 - V_{out}) = R_2 C_2 \frac{d^2 V_{out}}{dt^2}$$

Substituting this into the 1st equation yields:

$$R_1 C_1 R_2 C_2 \frac{d^2 V_{out}}{dt^2} + (R_1 + R_2) C_2 \frac{dV_{out}}{dt} + V_{out} = V_{in}$$

This 2nd order differential equation has a transient characteristic equation of:

$$R_1 C_1 R_2 C_2 s^2 + (R_1 + R_2) C_2 s + 1 = 0 \Rightarrow$$

$$s = \frac{-(R_1 + R_2) C_2 \pm \sqrt{[(R_1 + R_2) C_2]^2 - 4 R_1 C_1 R_2 C_2}}{2 R_1 C_1 R_2 C_2}$$

Thus, by carefully selecting component values, we can get real, complex, or repeated roots. Recall that these cases lead to overdamped, underdamped, and critically damped responses, respectively.

It is common to express the roots of the characteristic equation of a 2nd order circuit in the form $s = -\alpha \pm j \omega_d$ so that in the under-damped case (complex roots), the solution to the differential equation will be of the general form

$$V_{out}(t) = B_1 e^{-\alpha t} \cos(\omega_d t) + B_2 e^{-\alpha t} \sin(\omega_d t).$$

This is the form of a damped oscillation where α is the damping factor and ω_d is the damped radian frequency. Other common terms to describe a 2nd order circuit can be defined by expressing the roots of the characteristic equation in the form $s = -\alpha \pm j \sqrt{\omega_o^2 - \alpha^2}$, where ω_o is the resonant radian frequency of the circuit. This is the frequency that the circuit would oscillate at if it was completely undamped ($\alpha=0$). In that case, the circuit would be a simple harmonic oscillator that you previously studied (in physics). Finally it is common to refer to the quality factor or Q-factor of a 2nd order circuit which is defined as: $Q = \frac{\omega_o}{2\alpha}$

The Q-factor is a quantitative measure of the amount of damping in a 2nd order circuit. Note that the case of $\omega_o = \alpha$ will lead to repeated roots of the characteristic equation and a value of $Q = 1/2$. Thus, $Q = 1/2$ is the dividing line between underdamped and overdamped. For $Q > 1/2$ the roots are complex and the circuit is underdamped, while for $Q < 1/2$, the roots are real and the circuit is overdamped. Plots of typical output voltages for the underdamped and overdamped cases were shown in class (and are part of Lecture 8 notes). For an overdamped system, the real roots will cause the voltage response to have an exponential form, while in an underdamped system, the complex roots will produce a decaying sinusoidal response.

The textbook provides values for α , ω_o , and ω_d for the case of simple series and parallel RLC circuits. However, by using the formulation above, we can compute these quantities for more complicated 2nd order circuits like our Sallen-Key circuit. The following table summarizes these results:

	α	ω_0	ω_d	Q
Series RLC	$\frac{R}{2L}$	$\sqrt{\frac{1}{LC}}$	$\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$	$\sqrt{\frac{L}{R^2 C}}$
Parallel RLC	$\frac{1}{2RC}$	$\sqrt{\frac{1}{LC}}$	$\sqrt{\frac{1}{LC} - \left(\frac{1}{2RC}\right)^2}$	$\sqrt{\frac{R^2 C}{L}}$
Sallen-Key $\left(\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}\right)$	$\frac{1}{2R_{eq}C_1}$	$\sqrt{\frac{1}{R_1 C_1 R_2 C_2}}$	$\sqrt{\frac{1}{R_1 C_1 R_2 C_2} - \left(\frac{1}{2R_{eq}C_1}\right)^2}$	$\sqrt{\frac{C_1}{C_2} \frac{R_{eq}}{R_1 + R_2}}$

Prelab

Design a Sallen-Key circuit in LTSpice as shown above. Choose component values so that the circuit produces a critically damped response ($Q=1/2$) and a resonant radian frequency of $\omega_0 = 2000\pi$ rad/sec ($f_0 = 1$ kHz). Be sure to try and choose component values that are available to you in your lab kit. You will not be able to exactly achieve the design goals with the restrictions of the component values, but you should try to get as close as possible with what you have. Be sure to pick resistors in the $>1k\Omega$ range to allow the op-amp to drive the circuit successfully.

Repeat your design for each of the following cases:

- Over-damped
- Under-damped

Create a table with the component values for the three cases and bring this with you to the lab.

Using a 100Hz square wave with 2V peak-to-peak as your input source, run LTSpice simulations for each of the above cases. Print a graph of the transient response for each of the cases to submit with your lab report. Be sure to label your graphs!

Procedure

Parts and Equipment needed:

- Several resistors
- Several (non electrolytic) capacitors
- One 741 (or equivalent) op-amp
- XK700 Trainer
- Oscilloscope

Build and Measurements

- Build the Sallen-Key circuit you designed in the prelab using the component values from your table that produced a critically damped response.
- Use the function generator on the XK700 to produce a 100Hz square wave with 2V peak-to-peak for your input
- Display the input on CH1 and the output on CH2 of the oscilloscope
- Record the measured waveforms for your lab report.

Repeat the above steps for each of the five cases.

Lab Report

- Title page
- Procedure: Write in your own words what you did in lab
- Data and Results: For each of the three cases:
 - Theoretical: Solve the appropriate different equation (show your work)
 - Simulation: Provide your LTSpice simulation graph
 - Measured: Provide your measured results
- Discussion: Discuss any differences between theoretical, simulationa, and measured values. What changes could have been done to the procedure to make your results come out better if you had to do it over again?